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Checklist

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes]
 - (b) Complete proofs of all theoretical results. [Yes]
 - (c) Clear explanations of any assumptions. [Yes]
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes]
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. [Yes]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

A MATHEMATICAL PROOFS

Contents

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A.1 Proof of Lemma 3.2

In this section, we present the necessary and sufficient conditions required for achieving perfect concept erasure. First, we introduce the notion of principal inertia components, which will be instrumental in deriving the conditions for perfect erasure.

Principal Inertia Components (PICs). PICs can be viewed as a decomposition of the joint distribution, $p_{X,A}$, between two random variables X and A (a branch of study dedicated to this is called *correspondence analysis* (Greenacre, 2017)). Principal inertia components exhibit interesting properties and are often used to define common measures like maximum correlation, $\rho_m(X, A)$, and Chi-squared correlation, $\chi^2(X, A)$.

Definition A.1 (Principal Inertia Components (Calmon et al., 2017)). *Let random variables X and A have finite support sets $\mathcal{X}$ and $\mathcal{A}$ respectively, and joint distribution $p_{X,A}$. In addition, let $f_0 : X \rightarrow \mathbb{R}$ and $g_0 : Y \rightarrow \mathbb{R}$ be the constant functions $f_0(x) = 1$ and $g_0(a) = 1$. For $k \in \mathbb{Z}^+$, we (recursively) define:*

$$\lambda_k(X; A) = \max \left\{ \mathbb{E}[f(X)g(A)]^2 \mid \begin{array}{l} \mathbb{E}[f(X)f_j(X)] = 0, \\ \mathbb{E}[g(Y)g_j(A)] = 0, j \in \{0, \dots, k-1\} \end{array} \right\},$$

where $f \in \mathcal{L}_2(p(X))$, $g \in \mathcal{L}_2(p(Y))$ are square-integrable functions, and

$$(f_k, g_k) := \arg \max \left\{ \mathbb{E}[f(X)g(Y)]^2 \mid \begin{array}{l} \mathbb{E}[f(X)f_j(X)] = 0, \\ \mathbb{E}[g(Y)g_j(A)] = 0, j \in \{0, \dots, k-1\} \end{array} \right\}.$$

$\lambda_k(X; A)$ values are known as the principal inertia components (PICs) of the joint distribution, $p(X, A)$. The functions f_k and g_k are the principal functions of X and A .

The above definition presents a characterization of PICs as the construction of zero-mean, unit variance functions $f(X)$ and $g(A)$, which maximizes the correlation $\mathbb{E}[f(X)g(A)]$ without using any additional information. Note that alternative characterizations of PICs exist that include estimation errors and geometric interpretations (Calmon et al., 2017).

From Definition A.1, we observe that PICs satisfy: $\lambda_{k+1}(X; A) \leq \lambda_k(X; A) \leq 1$. PICs also correspond to the singular values of the joint probability matrix, specifically $\sqrt{\lambda_k(X; A)}$ is the $(k+1)$ -st singular value of $\mathbf{Q}$, where $\mathbf{Q} = \mathbf{D}_X^{-1/2} \mathbf{P} \mathbf{D}_A^{-1/2}$. $\mathbf{D}_X \in \mathbb{R}^{m \times m}$ and $\mathbf{D}_A \in \mathbb{R}^{n \times n}$ are diagonal matrices with entries $[\mathbf{D}_X]_{i,i} = p_X(i)$ and $[\mathbf{D}_A]_{j,j} = p_A(j)$ respectively, where $m = |\mathcal{X}|$ and $n = |\mathcal{A}|$. $\mathbf{P} \in \mathbb{R}^{m \times n}$ is a matrix with entries $[\mathbf{P}]_{i,j} = p_{X,A}(i, j)$.

k -correlation. Next, we introduce the notion of k -correlation and discuss how it relates to commonly known correlation metrics.

Definition A.2 (k -correlation). *For random variables X , A with finite support sets $\mathcal{X}$, $\mathcal{A}$ respectively, k -correlation between X and A is defined as:*

$$\mathcal{J}_k(X, A) := \sum_{i=1}^k \lambda_i(X, A). \quad (8)$$

The special cases of k -correlation correspond to well-known correlation metrics. For example, $\mathcal{J}_1(X, A) = \rho_m(X, A)^2$ is the Rényi maximum correlation (Buja, 1990). Similarly, if $d = \min\{|\mathcal{X}|, |\mathcal{A}|\} - 1$, then $\mathcal{J}_d(X, A) = \chi^2(X, A)$ is the chi-squared correlation.